

PURBANCHAL UNIVERSITY

2025

Master of Computer Application (M.C.A.) / First Semester / Final

Time: 03:00 hrs.

Full Marks: 60 / Pass Marks: 24

MCA103/MCA113: Operating System (New/Old Course)

Candidates are required to give their answers in their own words as far as practicable.

Group A

Answer TWO questions.

2×12=24

1. Differentiate between a process and a program. Explain any three process scheduling algorithms with suitable examples. 3+9
2. Why do we need page replacement algorithm? Explain FIFO, Optimal and LRU page replacement algorithm and calculate the total number of page faults for the following reference string, with 3 page frames.
Reference string: 3, 2, 5, 1, 3, 5, 2, 4, 7, 6, 4, 3, 2, 4, 6, 7, 5, 2, 7, 5
- 3(a) Define deadlock and explain deadlock prevention methods. 6
- (b) Explain the advantages and disadvantages of distributed operating system. 6

Group B

Answer SIX questions.

6×6=36

4. Discuss about the different types of operating system.
5. What is critical-section problem? Explain how this problem can be solved.
6. Explain internal and external fragmentation faced by swapping technique of memory management with suitable examples.
7. Explain file system implementation methods, with merits and demerits.
8. What is I/O software? Discuss about the importance of clock.
9. What do you mean by user-authentication? Briefly describe the file protection mechanisms.
10. What do you mean by virtual memory? Explain memory management using bitmaps.
11. Write short notes on Any TWO:
 - (a) Thread
 - (b) Sleep and wake-up
 - (c) Communication in distributed system

